

Honours Degree of Bachelor of Science in Information Technology & Management
Batch 23- Semester 4
CM 2111: Statistical Inference
Tutorial 02

01. Suppose you have a random sample $X_1, X_2, \dots, X_n$ from a Poisson distribution with an unknown parameter θ , ($X \sim Poi(\theta)$). Find the method of moment estimator of θ .

(Hint: $E(X) = \theta$)

02. Given a random sample $X_1, X_2, \dots, X_n$ from a Gamma distribution with unknown parameters α and β . Find the method of moment estimators for α and β .

(Hint: $E(X) = \alpha\beta$ and $Var(X) = \alpha\beta^2$).

03. Given a random sample $X_1, X_2, \dots, X_n$ from a Uniform distribution with unknown parameters a and b ($b > a$). Find the method of moment estimators for a and b .

(Hint: $E(X) = \frac{a+b}{2}$ and $Var(X) = \frac{(b-a)^2}{12}$)

04. Let X be a random variable with pdf

$$f(x; p) = \binom{m}{x} p^x (1-p)^{m-x} \quad ; x = 0, 1, 2, \dots, m$$

Find the Cramer-Rao lower bound for the variance of any unbiased estimator of p .

(Note that the range of X depends on m , but not on the unknown parameter p . Also, the sample size is $n = 1$).